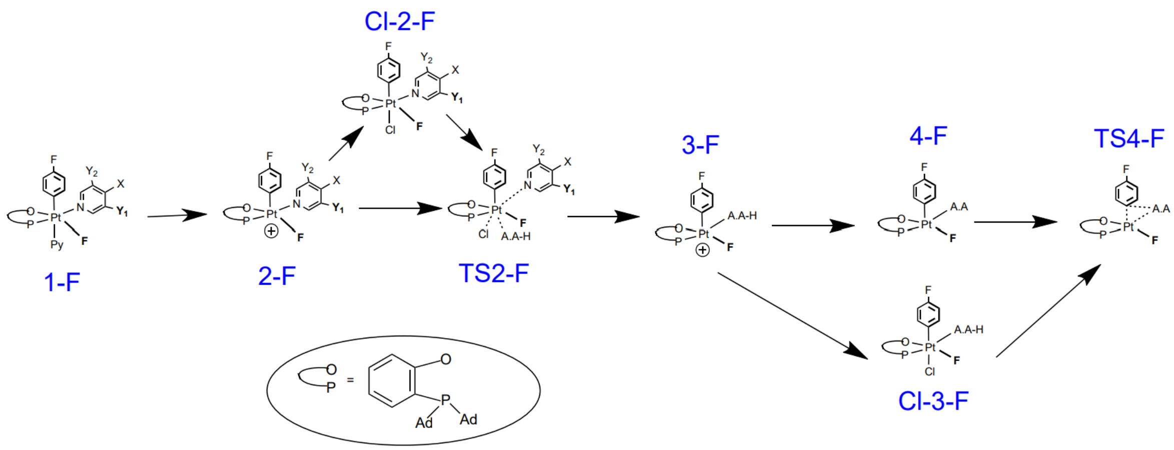
Modifying the Activation Barrier of Pt(IV)-Mediated Amino Acid and 4-Fluorophenyl Coupling through Pyridine Substituent Modification

Mahale Silva

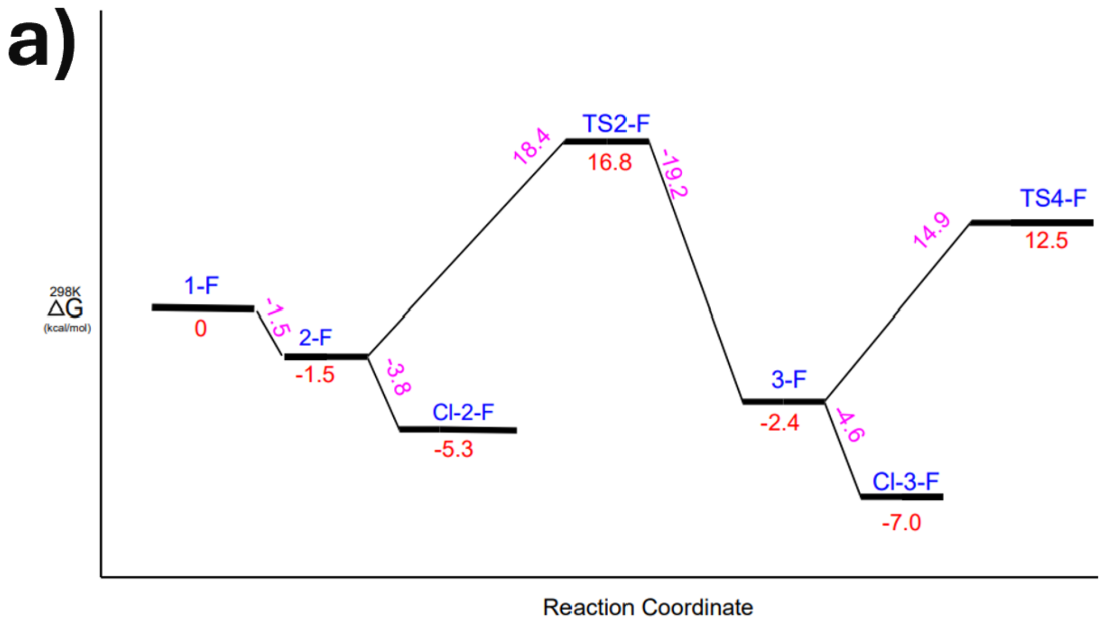
This work studies an octahedral Pt(IV) mediated C-N coupling of the terminal nitrogen of an amino acid (Glycine or Alanine) to 4-fluorophenyl. Following the mechanism in Scheme 1, The two studied transition states, **TS2-F** and **TS4-F**, are ligand substitution and C-N reductive coupling reactions, respectively. Following this study, substituents on the pyridine ligand can be placed at positions X, Y_1, and Y_2 to lower the activation barrier of the ligand substitution step.

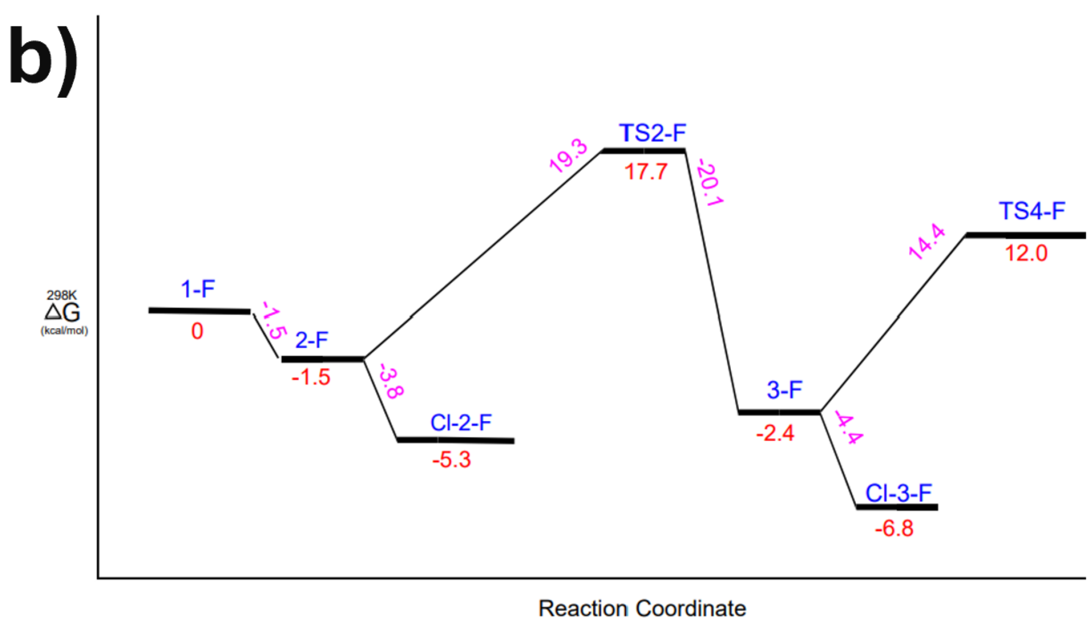


Scheme 1. Intermediates of energy profile of Pt(IV) aryl-fluoride complex. A.A. = glycine, L-alanine, or D-alanine. X = H, Cl, F, or CF_3. Y_1 and Y_2 = H or Cl. Representation of aryl group and O-P chelate ligand are shown.

The rate-determining step (RDS) was found using DFT calculations, where the specific C-N coupling step, **TS4-F**, was compared to the ligand substitution step seen in **TS2-F**. Following this, **TS2-F** with various substituents on the pyridine ligand were tested and compared to see which lowered the activation energy more. Although intermediate **4-F** is not focused on regarding the energy barrier analysis in the energy profile diagrams, it is significant in determining the orientation of the transition state since it is the reactant of this step. All calculations are done using the PBE functional and PCM solvation with DMSO.

Regarding the actual RDS of this pathway, it was determined that ligand substitution between **Cl-2-F** and **TS2-F** is the higher energy step compared to the C-N coupling step of **Cl-3-F** to **TS4-F** for glycine (22.1 kcal/mol vs. 19.5 kcal/mol), L-alanine (23.0 kcal/mol vs. 18.8 kcal/mol), and D-alanine (23.9 kcal/mol vs. 17.8 kcal/mol), as seen in Figure 1. Even by examining the actual step rather than the energy barrier after chlorine insertion (2-F to TS2-F, 3-F to TS4-F), the activation barrier of ligand substitution was still consistently higher. This result is likely because the substitution step involves the proper coordination and orientation of the amino acid when substituting itself with the pyridine ligand (amino acid coordinates from the site trans to the aryl ligand). Along with this, the steric congestion caused by this step is large, while C-N coupling reduces the intramolecular steric effects. Overall, the ligand substitution step sets up the favorable geometry of the C-N coupling step.





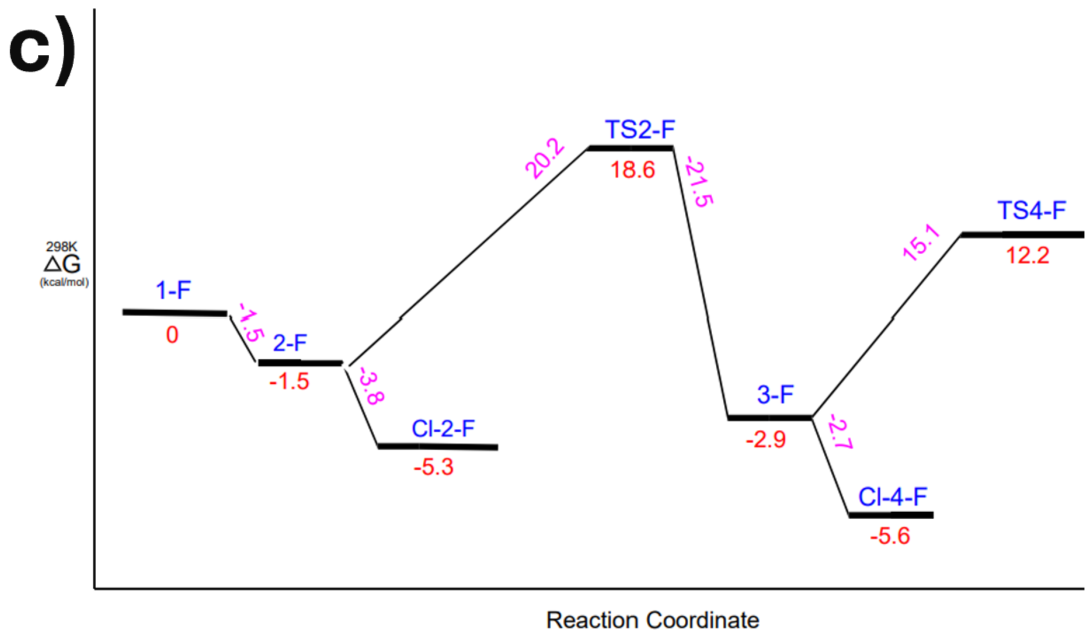


Figure 1. Energy profile diagrams (based off Scheme 1 pathway) for different mechanistic pathways of (a) glycine, (b) L-alanine, and (c) D-alanine. DFT-calculated Gibbs energies for intermediates calculated with standard state (298K and 1 atm) are in kcal mol^{-1} (TS refers to Transition State).

Following this, the substituents at multiple positions of the pyridine ligand involved in the ligand substitution, where para positions X= H, Cl, F, CF_3 and meta positions Y_1 (cis to F^- ligand) OR Y_2 (trans to F^- ligand) = Cl, were modified to decrease the barrier. The resulting energies can be seen in table 1 for the previously determined RDS.

Table 1. Resulting activation barriers between Cl-2-F and TS2-F for different systems of pyridine substituents with glycine, L-alanine, and D-alanine. Cis- and trans- refer to the steric relationship to the fluorine ligand of the platinum center.

	pyridine	p-chloropyridine	p-fluoropyridine	p-CF_3-pyridine	Cis-3-chloropyridine	Trans-3-chloropyridine
Glycine	22.1	21.4	22.0	21.9	21.2	20.5
L-Alanine	23.0	21.9	22.4	22.7	21.7	21.4
D-Alanine	23.9	22.8	23.3	23.3	22.3	22.2

For reference the Hammett constant, which quantifies the electron-donating or withdrawing properties of the substituents on the pyridine ligand relative to the unsubstituted pyridine with only hydrogen, for each of these substituents in Table 1 are 0, 0.23, 0.06, 0.54, 0.37, and 0.37, respectively, from left to right. The results show that trans-3-chloropyridine provides the lowest energy barrier, as it is slightly lower than its cis isomer but within uncertainty for DFT calculations. Comparisons of activation barriers with the Hammett constants show a deviation from the expected trend with the X=CF_3 substituent. This is likely because due to the high electron-withdrawing level of trifluoromethyl, which correlates with its high Hammett constant, pyridine becomes too weak of an electron donor to the metal center. Because of this, the chloride ligand intermediate is more stable than expected, increasing the activation barrier. When comparing between **2-F** and **TS2-F**, the substituents more closely followed the order of the Hammett constants as seen in Table 2, but the trifluoromethyl substituent is still higher in energy than expected due to the same reasoning.

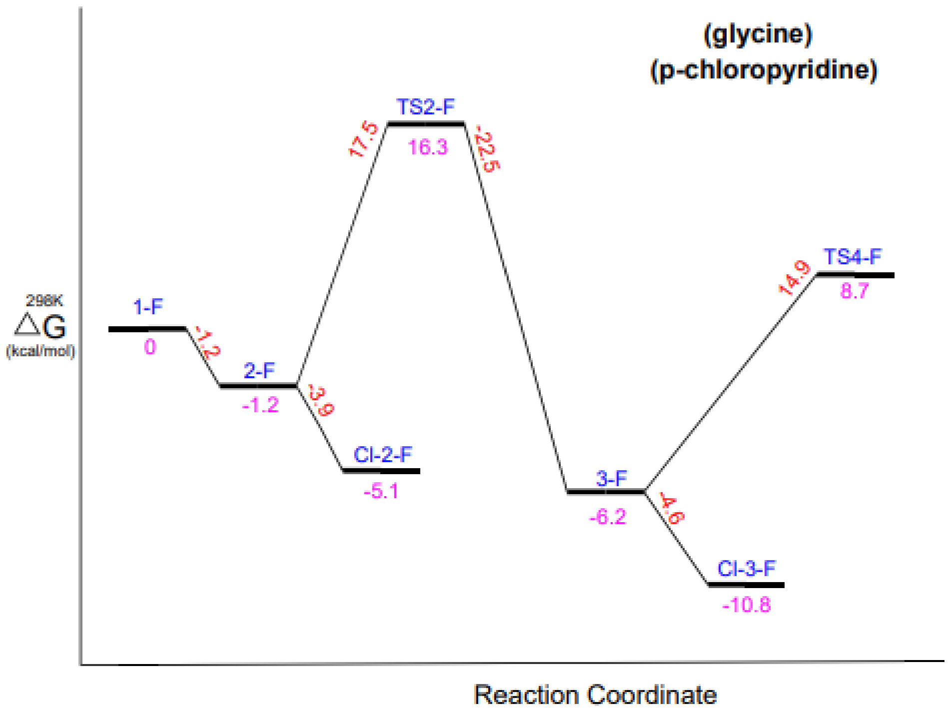
Table 2. Resulting activation barriers between 2-F and TS2-F for different systems of pyridine substituents with glycine, L-alanine, and D-alanine. Cis- and trans- refer to the steric relationship to the fluorine ligand of the platinum center.

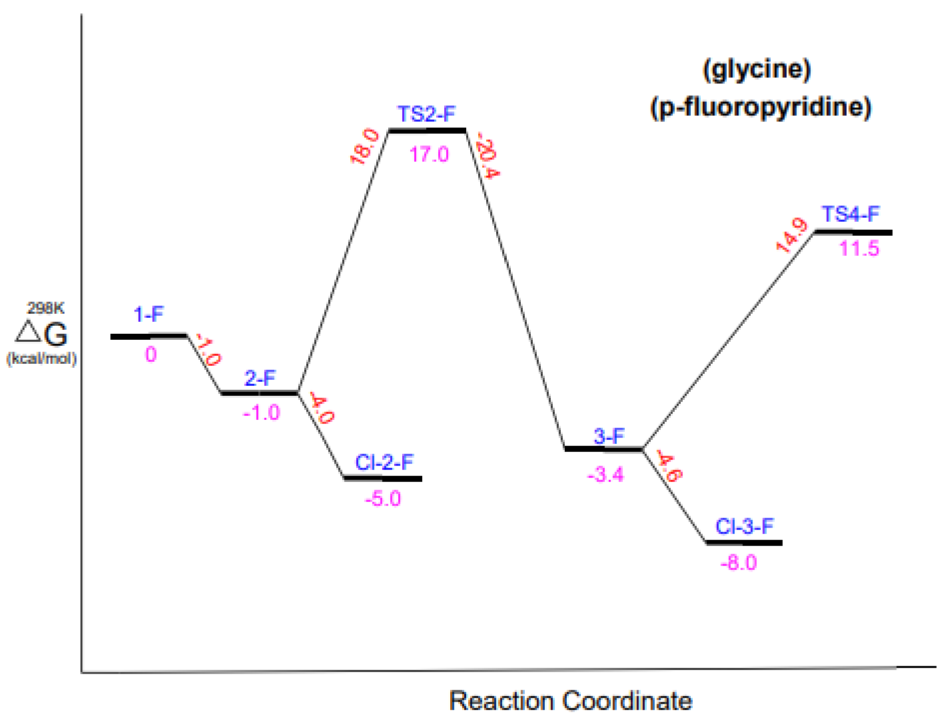
	pyridine	p-chloropyridine	p-fluoropyridine	p-CF_3-pyridine	Cis-3-chloropyridine	Trans-3-chloropyridine
Glycine	18.4	17.5	18.0	17.1	17.2	16.9
L-Alanine	19.3	18.0	18.4	17.9	17.7	17.7
D-Alanine	20.2	18.9	19.3	18.5	18.4	18.6

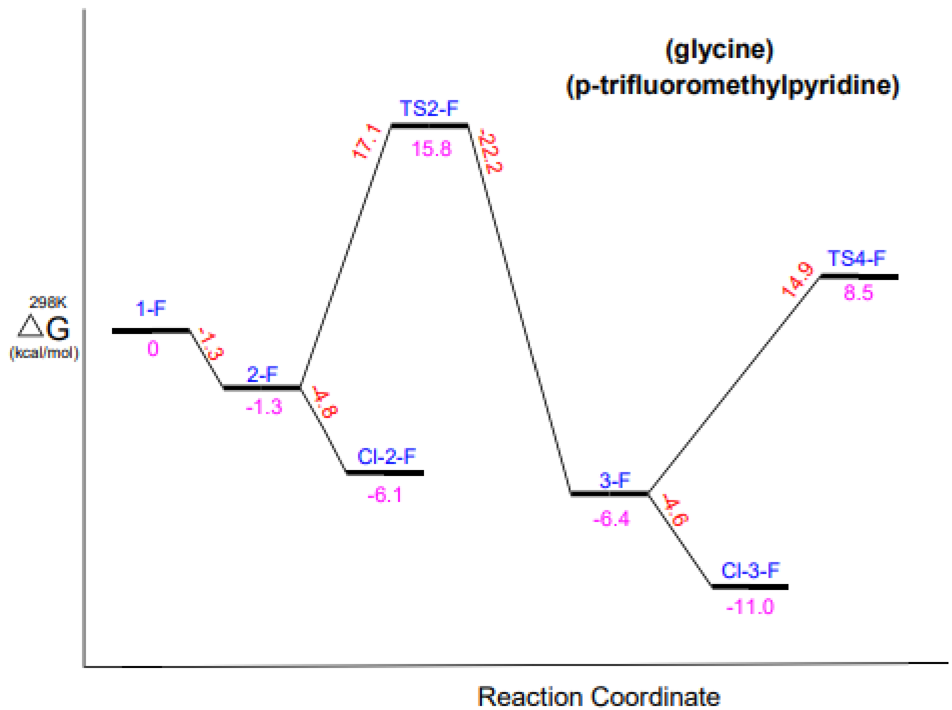
Overall, this suggests that the substituent electronics and Hammett constants do influence the barrier to an extent and that the larger deviation comes specifically from the chloride insertion intermediate.

Supplemental Information

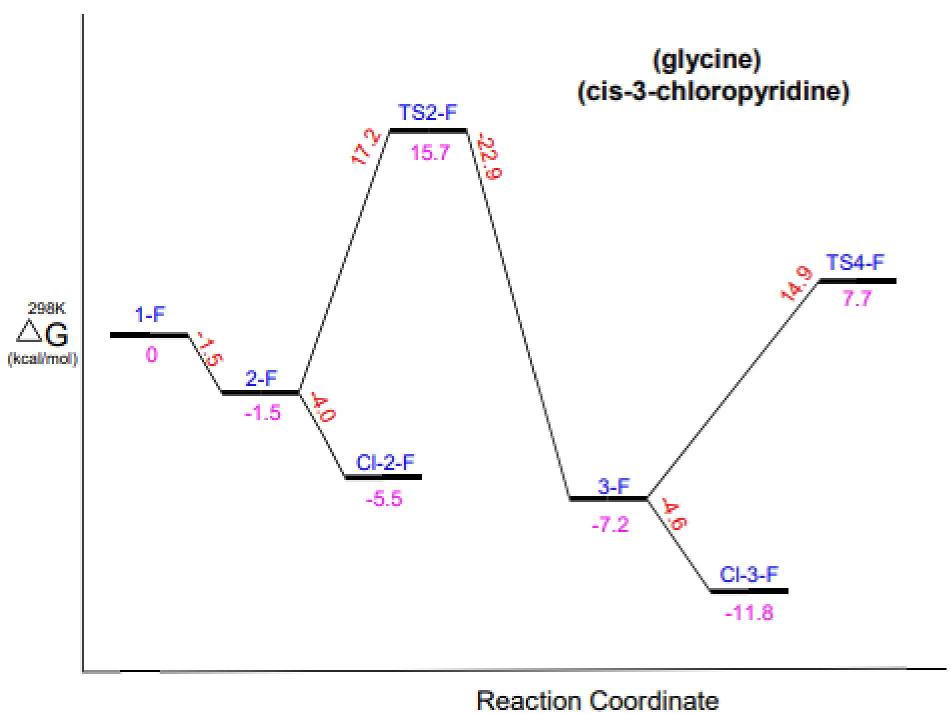
Figure 2. All labeled mechanistic pathways for all types of modified pyridine ligands for each amino acid. DFT-calculated Gibbs energies for intermediates calculated with standard state (298K and 1 atm) are in kcal mol^{-1} (TS refers to Transition State).







(glycine)
(p-trifluoromethylpyridine)
298K
△G
(kcal/mol)
1-F
0
-1.3
2-F
-1.3
-4.8
Cl-2-F
-6.1
17.1
TS2-F
15.8
-22.2
3-F
-6.4
-4.6
Cl-3-F
-11.0
14.9
TS4-F
8.5
Reaction Coordinate



(glycine)
(cis-3-chloropyridine)
298K
△G
(kcal/mol)
1-F
0
-1.5
2-F
-1.5
-4.0
Cl-2-F
-5.5
17.2
TS2-F
15.7
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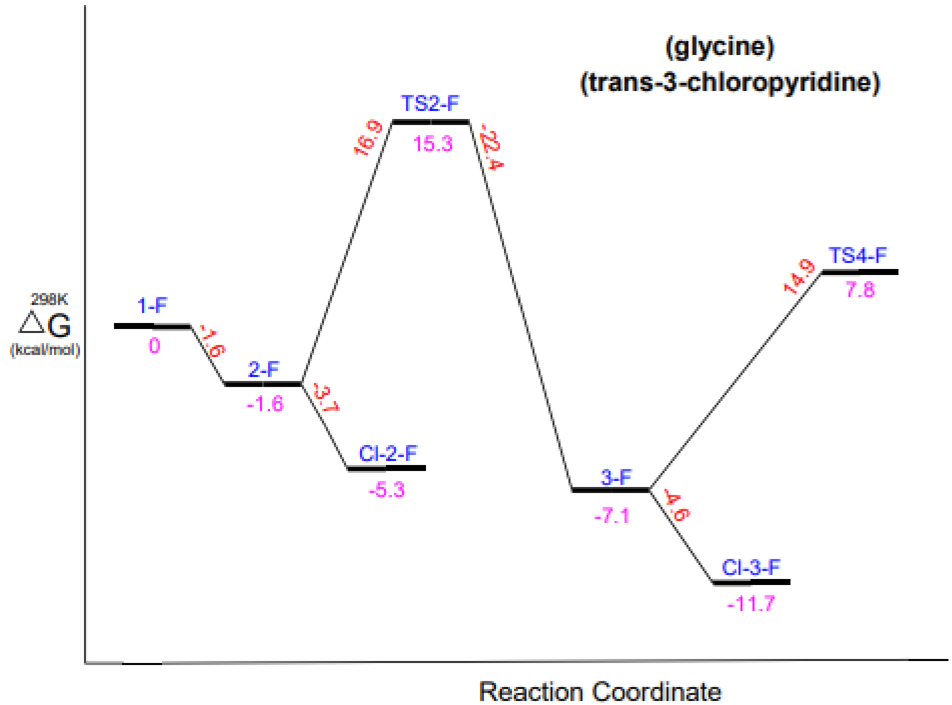
Cl-3-F
-11.8
14.9
TS4-F
7.7
Reaction Coordinate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(trans-3-chloropyridine)
298K
ΔG
(kcal/mol)
1-F
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2-F
-1.6
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Cl-2-F
-5.3
16.9
TS2-F
15.3
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Cl-3-F
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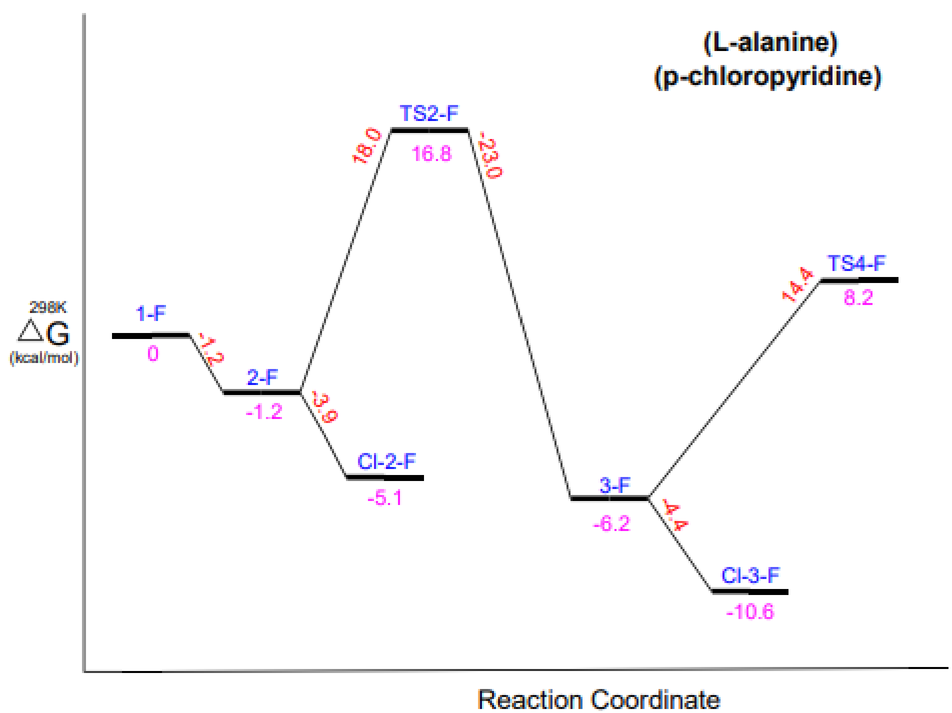
14.9
TS4-F
7.8
Reaction Coordinate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ΔG
(kcal/mol)
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-3.9
Cl-2-F
-5.1
18.0
TS2-F
16.8
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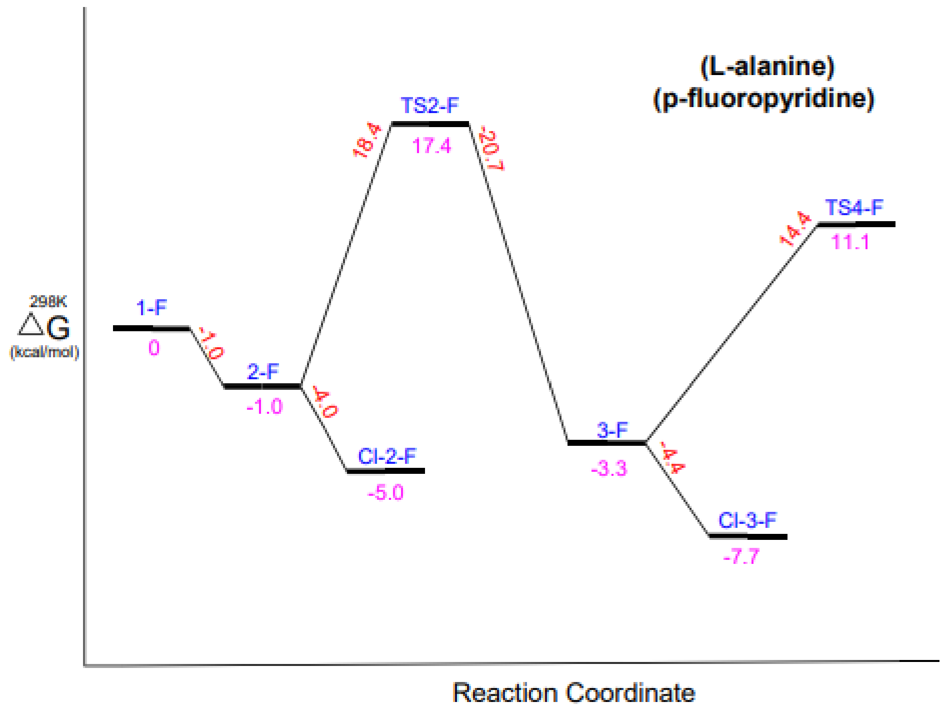
Reaction Coordinate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(p-fluoropyridine)
298K
ΔG
(kcal/mol)
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2-F
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-4.0
Cl-2-F
-5.0
18.4
TS2-F
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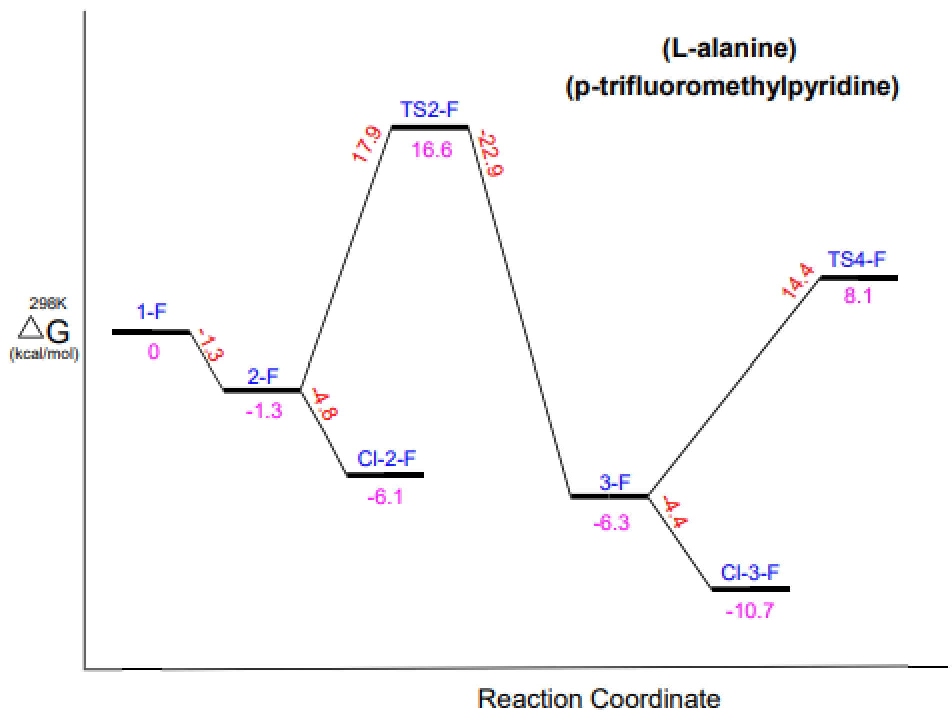
11.1
Reaction Coordinate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(p-trifluoromethylpyridine)
298K
ΔG
(kcal/mol)
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2-F
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Cl-2-F
-6.1
17.9
TS2-F
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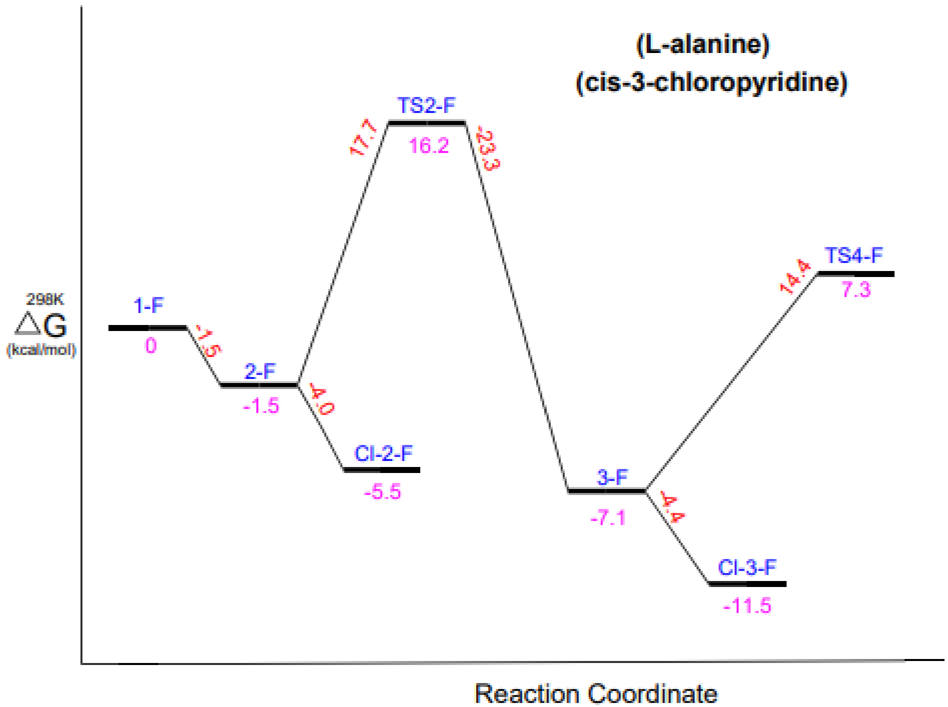

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(cis-3-chloropyridine)
298K
△G
(kcal/mol)
1-F
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-1.5
2-F
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-4.0
Cl-2-F
-5.5
17.7
TS2-F
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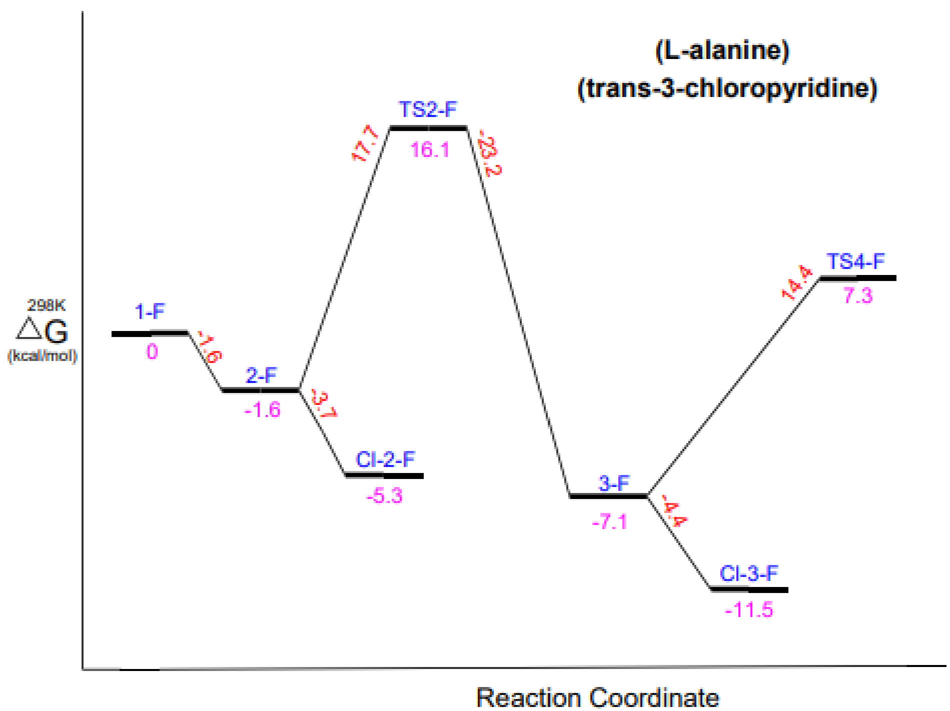

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(trans-3-chloropyridine)
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△G
(kcal/mol)
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2-F
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-3.7
Cl-2-F
-5.3
17.7
TS2-F
16.1
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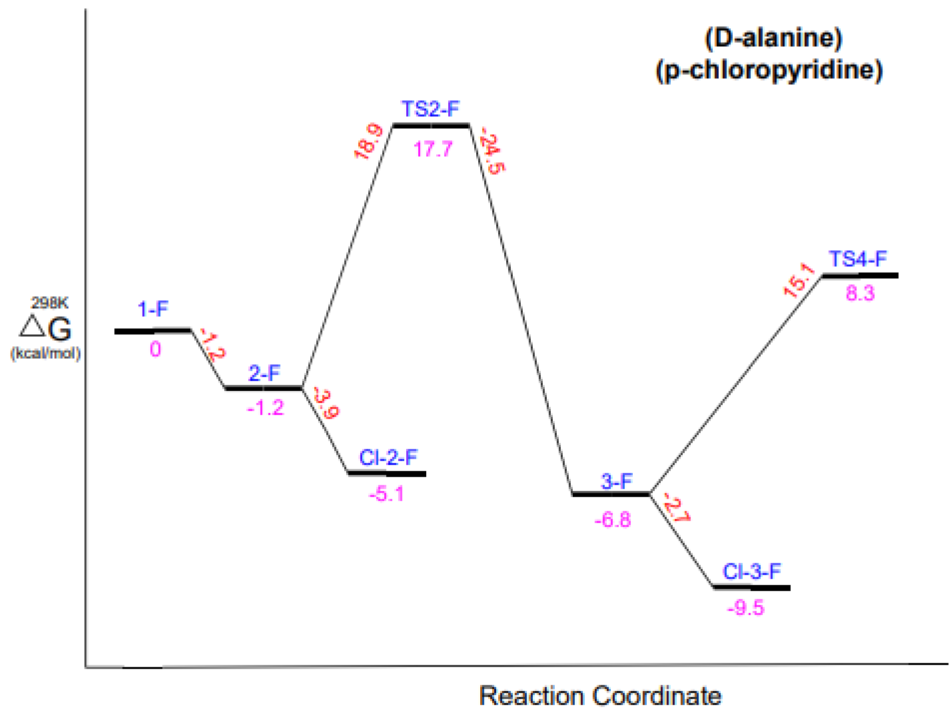
Reaction Coordinate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△G
(kcal/mol)
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18.9
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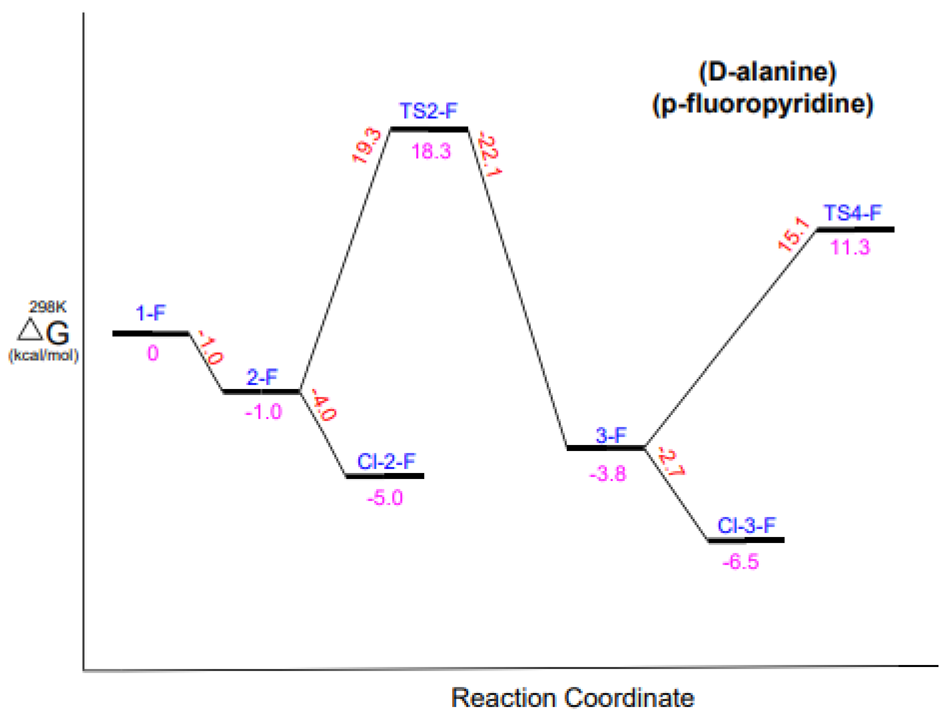
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(p-fluoropyridine)
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(kcal/mol)
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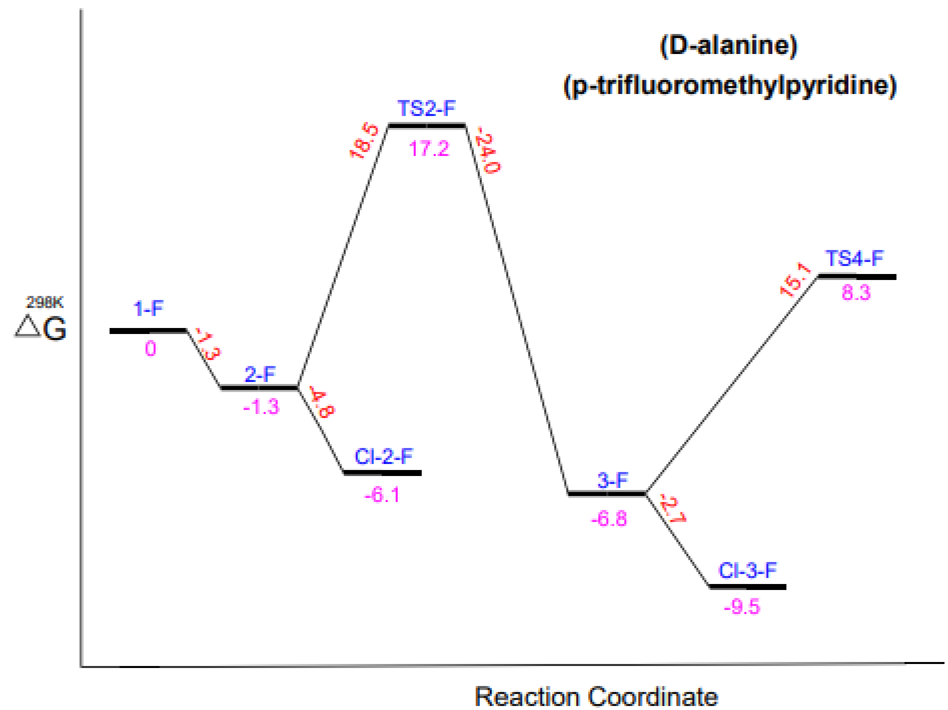
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(p-trifluoromethylpyridine)
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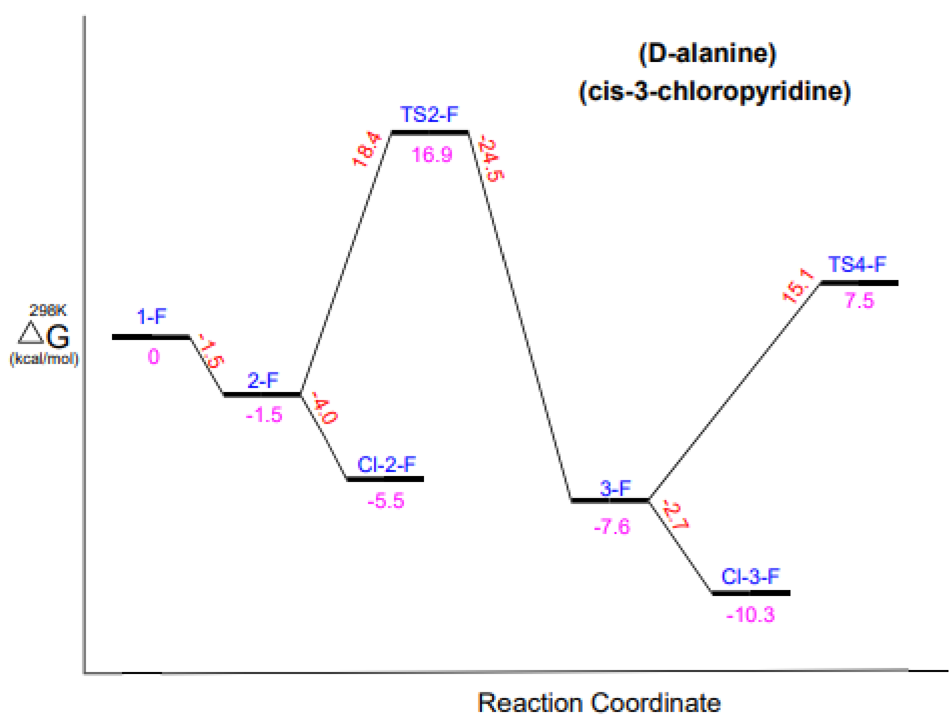
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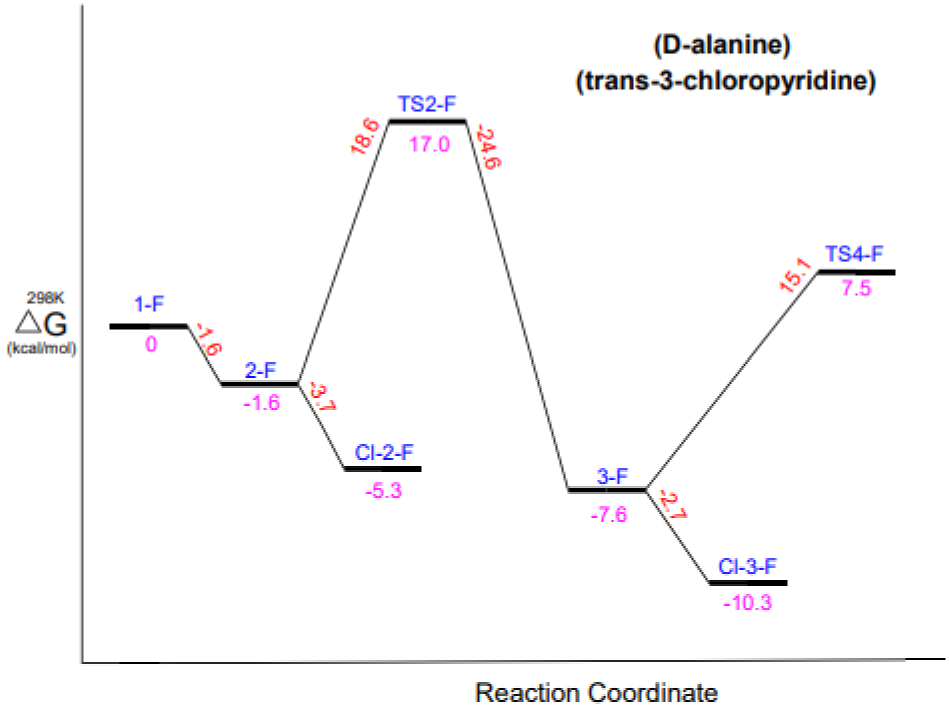
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